# Tic-Tac-Toe AI using Minimax Algorithm

## Project Report

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## 1 Introduction

Tic-Tac-Toe is a simple two-player game, but creating an AI that plays optimally requires strategic decision-making. In this project, an AI-based opponent is developed using the Minimax algorithm.

## 2 Problem Statement

The objective is to design an AI agent that can play Tic-Tac-Toe against a human player and make optimal decisions at every step.

## 3 Approach

The problem is treated as an adversarial search problem where:

- The human player tries to win
- The AI tries to maximize its chances of winning

The Minimax algorithm is used:

- The AI explores all possible future moves
- It assumes the opponent also plays optimally

- It selects the move with the best outcome

# 4 Implementation Details

The program includes:

- Board representation using a 2D list
- Functions to check game state (win/draw)
- Implementation of Minimax algorithm
- User interaction through command line

# 5 Features

- Human vs AI gameplay
- AI makes optimal decisions
- Detects win, loss, and draw conditions

# 6 Results

## 6.1 Game Start

```
You are X, AI is O

  |   |
--+---+--
  |   |
--+---+--
  |   |

Enter row (0-2):
```

```
Enter row (0-2): 1
Enter col (0-2): 2
  |   | O
--+---+--
  |   | X
--+---+--
  |   |

Enter row (0-2): 1
Enter col (0-2): 1
  |   | O
--+---+--
O | X | X
--+---+--
  |   |

Enter row (0-2): 0
Enter col (0-2): 1
  | X | O
--+---+--
O | X | X
--+---+--
  | O |

Enter row (0-2): 0
Enter col (0-2): 1
Cell already taken
Enter row (0-2): 0
Enter col (0-2): 0
X | X | O
--+---+--
O | X | X
--+---+--
  | O | O
```

### 6.2 Game Result

```
Enter row (0-2): 2
Enter col (0-2): 0
X | X | O
--+---+--
O | X | X
--+---+--
X | O | O

It's a draw!
```

## 7 Challenges Faced

Implementing the Minimax algorithm correctly was the main challenge. It was important to evaluate all possible game states and ensure the AI makes the best decision.

## 8 Conclusion

This project shows how adversarial search techniques can be used to build intelligent agents. The AI is able to make optimal decisions and never lose.

## 9 What I Learned

- Understanding of adversarial search
- How Minimax algorithm works

- How AI can simulate decision making